

M. C. N. REQUIRED
THIS PROGRAM REQUIRES MCN(S)
IN ORDER TO WORK PROPERLY

IDENTIFICATION

PRODUCT CODE: MAINDEC-Ø8-DHKLC-B-D
PRODUCT NAME: KL8F DOUBLE BUFFERED
ASYNCHRONOUS INTERFACE DIAGNOSTIC
DATE REVISED: MAY 1972
MAINTAINER: DIAGNOSTIC GROUP
AUTHOR: PATRICK COYNE

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1.0 ABSTRACT

THIS DIAGNOSTIC FACILITATES THE CHECK-OUT OF THE KL8F DOUBLE BUFFERED ASYNCHRONOUS INTERFACE. THIS IS A CLOSED LOOP TEST, A METHOD TO CONNECT EIA OUTPUT TO EIA INPUT IS REQUIRED, REFER TO TEST PROCEDURE M8652-0-3 FOR CONFIGURATION, ERROR HALTS AND SCOPE LOOPS ARE PROVIDED.

2.0 REQUIREMENTS

2.1 EQUIPMENT

POP-8E COMPUTER
ASR-33 TELETYPE OR EQUIVALENT DEVICE
M8652 QUAD MODULE
ONE LOOP BACK PLUG #7008517
IF LOOP BACK PLUG IS NOT AVAILABLE, CONNECT PINS
E TO M, AND F TO J ON CONNECTOR J1 OF M8652 MODULE.

2.2 STORAGE

THE PROGRAM OCCUPIES MEMORY LOCATIONS 0000 TO 3000.

3.0 LOADING PROCEDURE

LOAD PROGRAM VIA BINARY LOADER.

4.0 STARTING PROCEDURE

LOAD STARTING ADDRESS 0200-DEPRESS CONTINUE. PROGRAM WILL HALT AT LOCATION 0202.

FIRST PROGRAM HLT (0202) IS TO ALLOW OPERATOR TO SELECT IOT STRUCTURE, THROUGH THE SWITCHES, FOR WHICH HIS M8652 HAS BEEN JUMPERED TO OPERATE WITH.

SWITCHES	SELECTS
0-5	RECEIVE IOT
6-11	TRANSMIT IOT

FOR EXAMPLE, IF THE NUMBER 0304 WAS PLACED IN THE SWITCHES THE IOT STRUCTURE WOULD BE:

RECEIVE -	603X
TRANSMIT -	604X

WHERE X=0-7

DEPRESS CONTINUE
PROGRAM WILL HALT AT LOCATION 0204.

THE SECOND PROGRAM HLT (0204) IS TO ALLOW THE OPERATOR TO PLACE IN THE SWITCH REGISTER (S.I.R.) THE NUMBER OF DATA BITS PER CHARACTER TO BE TRANSMITTED.

THERE ARE FOUR POSSIBLE COMBINATIONS:

S.I.R.=0037	(5 DATA BITS)
S.I.R.=0077	(6 DATA BITS)
S.I.R.=0177	(7 DATA BITS)
S.I.R.=0377	(8 DATA BITS)

DEPRESS CONTINUE
PROGRAM WILL HALT AT LOCATION 0207.

THE THIRD PROGRAM HLT (0207) ALLOWS THE OPERATOR TO SELECT THROUGH THE S.R. THE TEST TO BE RUN, BAUD RATE AT WHICH DATA IS TO BE TRANSFERRED, AND THE TOTAL NUMBER OF BITS (INCLUDING START, STOP, PARITY) EACH CHARACTER IS COMPOSED OF, FOR CONTROL SWITCH SETTINGS REFER TO PARA. 5.1
DEPRESS CONTINUE PROGRAM WILL NOW HALT ONLY IF AN ERROR IS ENCOUNTERED;

5.0 OPERATING PROCEDURE

5.1 CONTROL SWITCH SETTINGS

S.R. BIT(S)	SET AS	ACTION ON PROGRAM
0	0	STAY IN SCOPE LOOP
1,2,3	1	EXIT SCOPE LOOP
	0	RUN ALL TESTS
	1	TEST ONE ONLY
	2	TEST TWO ONLY
	3	TEST THREE ONLY
	4	TEST FOUR ONLY
	5	TEST FIVE ONLY
	6	TEST SIX ONLY
	7	TEST SEVEN ONLY

NOT USED

7	7 BITS PER CHARACTER
8	8 BITS PER CHARACTER
9	9 BITS PER CHARACTER
10	10 BITS PER CHARACTER
11	11 BITS PER CHARACTER
12	12 BITS PER CHARACTER
NOT USED	
NOT USED	

110	BAUD
134.5	BAUD
150	BAUD
300	BAUD
600	BAUD
1200	BAUD
1800	BAUD
2400	BAUD

NOTE(A): USED TO SELECT TOTAL NUMBER OF BITS PER CHARACTER, INCLUDING DATA (5,6,7 OR 8), START (1), STOP (1 OR 2), PARITY (0,1).

NOTE(B): USED TO SELECT BAUD RATE AT WHICH DATA IS TRANSFERRED.

5.2 THE OPERATOR HAS THE OPTION OF RUNNING ALL TESTS OR ANY ONE TEST, THROUGH THE SETTING OF THE CONTROL SWITCHES, REFER TO PARA 5.1.1.

5.3 AS AN INDICATION THAT A TEST HAS RUN SUCCESSFULLY THE M.O. REGISTER IS LOADED WITH THE TEST NUMBER (1,2,3,4,5,6,7) AFTER IT HAS MADE A COMPLETE PASS.

6.0 ERRORS

UPON DETECTION OF AN ERROR DURING ANY TEST THE PROGRAM WILL HALT. FOR A DESCRIPTION OF EACH ERROR REFER TO THE PROGRAM LISTING.

6.1 SCOPE LOOPS - ERROR RECOVERY

SCOPE LOOPS ARE PROVIDED FOR ALL ERRORS. TO ENTER SCOPE LOOP AFTER ENCOUNTERING AN ERROR HALT, DEPRESS KEY CONTINUE. ALL SCOPE LOOPS MAY BE EXITED BY PUTTING S.R. 0 TO A ONE.

DUE TO TIMING CONSIDERATIONS TWO DIFFERENT TYPES OF SCOPE LOOP ARE USED. ALL TIMING AND CONTROL TESTS (TST1, TST2, TST3, TST4, TST5, AND TST7A) SCOPE LOOPS WHEN EXITED (SW000) WILL GO TO THE NEXT TEST OR SUBTEST IN SEQUENCE. FOR EXAMPLE, WHEN EXITING A SCOPE LOOP IN TST3A, THE PROGRAM WOULD THEN CONTINUE ON TO TST3B.

ON THE OTHER HAND ALL DATA PATTERN TEST (TST6, TST7B) SCOPE LOOPS WHEN EXITED WILL RETURN TO THE HLT (0207) WHICH REQUIRES THE OPERATOR TO PLACE IN THE S.R. THE CONTROL SWITCH SETTING. AT THIS TIME HE MAY SELECT ANOTHER TEST AND CONTINUE.

7.0 RESTRICTIONS

STARTING RESTRICTIONS

THE OPERATOR HAS THE CHOICE OF THREE RESTART LOCATIONS, RESTARTING AT 0200 WILL NECESSITATE SELECTING IOT STRUCTURE, NUMBER OF DATA BITS PER CHARACTER, AND CONTROL SWITCH SETTINGS. THE SECOND RESTART ADDRESS WOULD BE LOCATION 0203, THE OPERATOR WOULD LOAD ADDRESS 0203 AND THEN SETUP THE NUMBER OF DATA BITS PER CHARACTER BEFORE DEPRESSING CONTINUE. THE PROGRAM WOULD THEN HALT FOR CONTROL SWITCH SETTINGS. THE THIRD RESTART LOCATION WOULD BE ADDRESS 0210. THE OPERATOR WOULD LOAD ADDRESS 0210 AND THEN SETUP THE CONTROL SWITCH SETTINGS BEFORE DEPRESSING CONTINUE.

PROGRAM DESCRIPTION

THE FIRST FUNCTION PERFORMED BY THE PROGRAM IS TO DETERMINE, THROUGH THE USE OF THE THREE PROGRAM HALTS PREVIOUSLY DESCRIBED (REFER TO PARA 4.0), THE CONFIGURATION OF THE M8652 MODULE REGARDING, IOT STRUCTURE, BIT CONFIGURATION AND BAUD RATE.

8.1 THIS PROGRAM CONTAINS SEVEN TESTS:

TST1 - TRANSMIT CONTROL LOGIC TEST
TST2 - TRANSMIT TIMING TEST
TST3 - RECEIVE CONTROL LOGIC TEST
TST4 - RECEIVE TIMING TEST
TST5 - BREAK TEST
TST6 - DATA TEST
TST7 - STATUS REGISTER TEST

8.2 TST1 - TRANSMIT CONTROL LOGIC TEST

THIS IS A BASIC TEST OF THE TRANSMITTER LOGIC, FUNCTIONS SUCH AS SKIPS, CLEARS, AND SETS ARE CHECKED. NO TIMING IS TAKEN INTO ACCOUNT AT THIS POINT.

8.3 TST2 - TRANSMIT TIMING TEST

THIS TEST BEGINS BY INSURING THAT THE TRANSMIT FLAG CAN BE SET AT A TIME APPROXIMATELY DOUBLE THAT OF THE BAUD RATE SELECTED (TST2A). THEN A TIME MUCH LOWER THAN THE BAUD RATE SELECTED IS USED TO VERIFY THAT THE FLAG IS NOT SETTING TOO SOON, AND FINALLY THE FLAG IS CHECKED AT THE CORRECT BAUD RATE.

8.4 TST3 - RECEIVE CONTROL LOGIC TEST

THIS IS A BASIC TEST OF THE RECEIVER LOGIC, BASIC COMMANDS ARE TESTED SUCH AS SKIPS, CLEARS, AND SETS, NO TIMING OR CHECKING OF DATA TRANSFERS IS PERFORMED.

8.5 TST4 - RECEIVE TIMING TEST

IS SIMILAR IN FUNCTION TO THE TRANSMIT TIMING TEST, IN THAT GROSS TIMES ARE FIRST USED TO VERIFY THAT THE RECEIVE FLAG CAN BE SET AND CLEARED, ONCE THIS IS VERIFIED THE CORRECT BAUD RATE IS THEN TESTED.

8.6 TST5 - BREAK TEST

CHECKS THE ABILITY OF UTPK TO GENERATE A BREAK AND AFTER APPROXIMATELY 235 MILS TO SET THE TRANSMIT FLAG, FLAG IS ALSO CHECKED FOR SETTING TOO SOON AND TOO LATE.

8.7 TST6 - DATA TEST

THIS TEST IS DIVIDED INTO TWO SECTIONS, TST6A WHICH GENERATES AND TRANSMITS A BINARY COUNT PATTERN, AND TST6B WHICH GENERATES AND TRANSMITS A RANDOM DATA PATTERN. EACH SECTION OF THIS TEST WHEN IT ENCOUNTERS AN ERROR WILL HALT WITH THE BAD DATA PATTERN IN THE AC. DEPRESSING KEY CONTINUE WILL STEP THE PROGRAM TO A SECOND HALT WHICH WILL DISPLAY THE GOOD DATA PATTERN IN THE AC. DEPRESSING KEY CONTINUE A SECOND TIME WILL PUT THE PROGRAM INTO A SCOPE LOOP. REFER TO PARA. 6.1 FOR INFORMATION REGARDING THE EXITING OF A DATA TEST SCOPE LOOP.

8.8 TST7 - STATUS REGISTER TEST

MADE UP OF TWO SUBTESTS TST7A AND TST7B. TST7A FORCES AN OVERRUN ERROR AND CHECKS THAT THE OVERRUN ERROR BIT (AC02) COMES UP AND COMBINED WITH STATUS WORD ENABLE WILL CAUSE THE STATUS ERROR BIT (AC00) TO COME UP. ERROR HALTS AND SCOPE LOOPS ARE PROVIDED, EXITING SCOPE LOOP, PROGRAM CONTINUES ON TO TST7B.

TST7B GENERATES A RANDOM DATA PATTERN AND CHECKS FOR STATUS ERRORS (OVERRUN, PARITY AND FRAMEING). WHEN AN ERROR OCCURS PROGRAM WILL HALT WITH FAILING STATUS BIT AND DATA WORD IN THE AC. SCOPE LOOP IS ENTERED BY DEPRESSING KEY CONTINUE. EXITING SCOPE LOOP IS THE SAME AS THAT FOR TST6, REFER TO PARA. 6.1.

FOR FURTHER INFORMATION REGARDING TESTS AND ERROR HALTS, REFER TO THE PROGRAM LISTING.

9.0 LISTING

//KL8F DOUBLE BUFFERED ASYNCHRONOUS INTERFACE
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//INSTRUCTION EQUALITIES//

4473	UKCF=JMS I	XXKCF	/CLEAR RECEIVE FLAG,
4474	UKSF=JMS I	XXKSF	/SKIP ON RECEIVE FLAG,
4475	UKCC=JMS I	XXKCC	/CLEAR RECEIVE FLAG AND AC,
4476	UKRS=JMS I	XXKRS	/INPUT DATA BUFFER V AC4-11 TO AC4-11
			/PARITY ERROR AND SWE V AC01 TO AC01
			/OVERRUN ERROR AND SWE V AC02 TO AC02,
			/FRAMING ERROR AND SWE V AC03 TO AC03,
			/(PE + OE + FE) AND SWE V AC00 TO AC00;
4477	UKIE=JMS I	XXKIE	/DATA 11 TO INTERRUPT ENABLE,
4500	UKRB=JMS I	XXKRB	/DATA 10 TO STATUS WORD ENABLE,
			/CLEAR AC AND RECEIVE FLAG,
4501	USPF=JMS I	XXSPF	/INPUT DATA BUFFER TO AC4-11,
4502	UTSF=JMS I	XXTSF	/SET TRANSMIT FLAG,
4503	UTCf=JMS I	XXTCF	/SKIP ON TRANSMIT FLAG,
4504	UIPC=JMS I	XXTPC	/CLEAR TRANSMIT FLAG,
4505	USPI=JMS I	XXSPI	/AC4-11 TO OUTPUT DATA BUFFER,
4506	UTLS=JMS I	XXTLS	/TRANSMIT, SET TRANSMIT FLAG WHEN DONE;
4507	UTPK=JMS I	XXTPK	/SKIP ON INTERRUPT FLAG,
			/CLEAR TRANSMIT FLAG,
			/AC4-11 TO OUTPUT DATA BUFFER,
			/TRANSMIT, SET TRANSMIT FLAG WHEN DONE;
			/GENERATE BREAK.

/CLEAR ALL FLAGS;
 /LOAD MQ,

6007	CAF=6007		
7421	MQL=7421		
4020	DELAY=JMS I	DELLAY	
4544	ERROR=JMS I	XFAIL	
4545	DATERR=JMS I	XSCOPE	
4043	SETCNT=JMS	COUNT	
4510	SET1A=JMS I	XSET1A	
4511	SET1B=JMS I	XSET1B	
4512	SET1C=JMS I	XSET1C	
4513	SET2A=JMS I	XSET2A	
4514	SET2B=JMS I	XSET2B	
4515	SET3A=JMS I	XSET3A	
4516	SET3B=JMS I	XSET3B	
4517	SET3C=JMS I	XSET3C	
4520	SET3D=JMS I	XSET3D	
4521	SET4A=JMS I	XSET4A	
4522	SET5A=JMS I	XSET5A	
4523	SET7A=JMS I	XSET7A	

//ERROR HALT DESCRIPTIONS//

1207	*HALT1	/SPF FAILED TO SET TRANSMIT FLAG OR
		/TSF FAILED TO SKIP ON TRANSMIT FLAG,
1207	HALT2	/CAF FAILED TO CLEAR TRANSMIT FLAG OR

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1210 1226    HALT3
/TSF SKIPPED ON TRANSMIT FLAG = 0.

1211 1242    HALT4
/TCF FAILED TO CLEAR TRANSMIT FLAG OR
/TSF SKIPPED ON TRANSMIT FLAG = 0.

1212 1256    HALT5
/CAF FAILED TO CLEAR ALL FLAGS OR
/SP1 SKIPPED WITHOUT INTERRUPT REQUEST OR
/AN ILLEGAL INTERRUPT DID OCCUR.

1213 1274    HALT6
/TRANSMIT FLAG FAILED TO CAUSE INTERRUPT,
/SP1 FAILED TO SKP ON INTERRUPT REQUEST OR
/SPF FAILED TO SET TRANSMIT FLAG OR
/TRANSMIT FLAG DID NOT SET INTERRUPT REQUEST.

1214 1310    HALT7
/CAF FAILED TO CLEAR ALL FLAGS OR
/AN ILLEGAL INTERRUPT TOOK PLACE.

1215 1413    HALT8
/TPC FAILED TO SET TRANSMIT FLAG OR
/FLAG TAKING TOO LONG TO SET.

1216 1424    HALT9
/TL5 FAILED TO CLEAR TRANSMIT FLAG OR
/TRANSMIT FLAG SETTING TOO SOON OR
/TSF SKIPPED ON TRANSMIT FLAG = 0.

1217 1434    HALT10
/TL5 FAILED TO SET TRANSMIT FLAG OR
/TRANSMIT FLAG TAKING TOO LONG TO SET OR
/TSF FAILED TO SKIP ON TRANSMIT FLAG.

1220 1461    HALT11
/TRANSMIT FLAG SETTING TOO SOON OR
/TSF SKIPPED ON TRANSMIT FLAG = 0.

1221 1471    HALT12
/TRANSMIT FLAG TAKING TOO LONG TO SET.

1222 1513    HALT13
/TL5 FAILED TO SET TRANSMIT FLAG OR
/FLAG IS BEING SET TOO LATE.

1223 1606    HALT14
/KCC FAILED TO CLEAR AC.

1224 1634    HALT15
/KSF FAILED TO SKIP OR
/RECEIVE WAS NOT SET OR WAS SET TOO LATE.

1225 1643    HALT16
/KCC FAILED TO CLEAR RECEIVE FLAG OR
/KSF SKIPPED ON RECEIVE FLAG = 0.

1226 1656    HALT17
/RECEIVE FLAG FAILED TO SET OR SET TOO LATE
/OR KSF FAILED TO SKIP.

1227 1665    HALT18
/KCF FAILED TO CLEAR RECEIVE FLAG
/OR KSF SKIPPED ON RECEIVE FLAG = 0.

1230 1703    HALT19
/KIE FAILED TO DISABLE INTERRUPT ENABLE F/F
/OR SP1 SKIPPED ON INTERRUPT ENABLE = 0.

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1231 1715 HALT20 /K1E AND AC11#1 FAILED TO ENABLE INTERRUPT ENABLE F/F
/OR SPI FAILED TO SKIP;

1232 1744 HALT21 /INTERRUPT DID NOT TAKE PLACE;

1233 2012 HALT22 /RECEIVE FLAG SET TOO SOON
/OR KSF SKIPPED ON RECEIVE FLAG = 0.

1234 2022 HALT23 /RECEIVE FLAG NOT SET OR SET TOO LATE
/OR KSF FAILED TO SKIP;

1235 2045 HALT24 /RECEIVE FLAG FAILED TO SET OR SET TOO LATE,

1236 2215 HALT25 /TPK FAILED TO SET TRANSMIT FLAG,

1237 2231 HALT26 /TRANSMIT FLAG SET TOO SOON,

1240 2246 HALT27 /TRANSMIT FLAG SET TOO LATE OR DID NOT SET;

1241 2256 HALT28 /TCF FAILED TO CLEAR TRANSMIT AND/OR
/BREAK FLAGS;

1242 2442 HALT29 //BINARY COUNT PATTERN FAILURES//

1243 2446 HALT30 /INCORRECT DATA DISPLAYED IN AC,

1244 2506 HALT31 /CORRECT DATA DISPLAYED IN AC,

1245 2511 HALT32 /INCORRECT DATA DISPLAYED IN AC,

1246 2631 HALT33 /CORRECT DATA DISPLAYED IN AC,

1247 2642 HALT34 //STATUS TEST ERRORS//

1250 2710 HALT35 /OVERRUN ERROR DID NOT OCCUR,

1251 2720 HALT36 /BIT 0 FAILED TO GO TO A ONE ON OVERRUN ERROR,

1252 2730 HALT37 /PARITY ERROR, FAILING STATUS BIT AND DATA WORD TO AC,

1253 2740 HALT38 /OVERRUN ERROR, FAILING STATUS BIT AND DATA WORD TO AC,

0001 0002 *0001 /FRAMEING ERROR, FAILING STATUS BIT AND DATA WORD TO AC,
5402 RETURN, 0 /ERROR BIT (AC0) = 1;
0002

JMP 1 RETURN

0020 0021 *20
0022 DELAY, 0 CLA CLL

```

2022	1124	TAD	NDELAY
0023	3125	DCA	DELAYN
0024	1126	TAD	CON100
0025	3127	DCA	US122
0026	2127	ISZ	US100
0027	5026	JMP	,=1
0030	7200	CLA	
0031	7200	CLA	
0032	7200	CLA	
0033	7200	CLA	
0034	2125	ISZ	DELAYN
0035	7610	SKP	CL A
0036	5420	JMP	I
0037	0420	AND	I
0040	0420	AND	I
0041	0020	AND	DELAY
0042	5024	JMP	DELOOP

```
//ROUTINE TO SETUP COUNTER FOR TEST LOOPS//
```

//POINTER FOR EXACT BAUD RATE TABLES//

0053	3000	ONBAUD,	ON110
0054	3006		ON134
0055	3014		ON150
0056	3022		ON300
0057	3030		ON600
0060	3036		ON1200
0061	3044		ON1800
0062	3052		ON2400

```
//POINTER FOR LOW END VALUE BAUD RATE TABLES//
```

0063	3060	LOBAUD,	L0110
0064	3066		L0134
0065	3074		L0150
0066	3102		L0300
0067	3110		L0600
0070	3116		L01200
0071	3124		L01800
0072	3132		L02400

```
//POINTERS FOR IOT SUBROUTINES//
```

0073	0600	XXCF,	XXCF
0074	0604	XXSF,	XXSF

2275	0612	XXKCC,	XXCC
2276	0616	XXKRS,	XXRS
2277	0622	XXKIE,	XXIE
2100	0626	XXKRB,	XXRB
2101	0632	XXSPF,	XSPF
2102	0636	XXTSF,	XTSF
2103	0644	XXTCF,	XTCF
2104	0650	XXTPC,	XTPC
2105	0654	XXSPI,	XSPI
0106	0662	XXTLS,	XTLS
0107	0666	XXTPK,	XTPK

//LINKS FOR ERROR HALT ROUTINES//

0110	1000	XSET1A,	SETT1A
0111	1010	XSET1B,	SETT1B
0112	1016	XSET1C,	SETT1C
0113	1024	XSET2A,	SETT2A
0114	1034	XSET2B,	SETT2B
0115	1044	XSET3A,	SETT3A
0116	1050	XSET3B,	SETT3B
0117	1062	XSET3C,	SETT3C
0120	1070	XSET3D,	SETT3D
0121	1074	XSET4A,	SETT4A
0122	1104	XSET5A,	SETT5A
0123	1116	XSET7A,	SETT7A

//CONSTANTS AND WORK LOCATIONS//

0124	0000	NDELAY,	0
0125	0000	DELAYN,	0
0126	7754	CON100,	-24
0127	0000	US100,	0
0130	0000	GETBAK,	0
0131	0000	NXTST,	0
0132	0000	TSTCNT,	0
0133	0000	XMTCH,	0
0134	0000	BITMSK,	0
0135	1233	RP1B,	1233
0136	7622	RP2B,	7622
0137	0000	HOLD1,	0
0140	0000	TEMP1,	0
0141	0000	TNOW,	0
0142	0000	TOLD,	0
0143	0000	TLAST,	0
0144	0312	XFAIL,	FAIL
0145	0332	XSCOPE,	SCOPE
0146	0000	BAUDNO,	0

0200	0200	*200	
0200	7300	BEGIN,	CLA CLL
0201	6007	CAF	

2232	7402	HLT	SETIOT	/GET SWITCHES FOR IOT/S.
2233	4777	JMS		
2234	7402	HLT		/GET SWITCHES TO DETERMINE
2235	7604	LAS		
2236	3134	DCA	BITMSK	/NUMBER OF DATA BITS PER WORD.
2237	7402	HLT		/CHECK SWITCH SETTINGS.
2238	4250	JMS	DECODE	/GET SETTING OF SW'S 9,10,11.
2239	1146	TAD	BAUDNO	/ADD POINTER OF BAUD TABLE.
2240	1376	TAD	(ONBAUD	
2241	3140	DCA	TEMP1	/SAVE NEW POINTER.
2242	1540	TAD I	TEMP1	/GET BAUD RATE THRU POINTER.
2243	1275	TAD	BITNO	/ADD BIT SIZE OF CHARACTER.
2244	3140	DCA	TEMP1	/SAVE FINAL POINTER.
2245	1540	TAD I	TEMP1	/PICK-UP BAUD RATE CONSTANT
2246	3277	DCA	ONRATE	/SAVE IT.
2247	1277	TAD	ONRATE	
2248	1277	TAD	ONRATE	/DOUBLE RATE.
2249	3301	DCA	DOUBLE	/SAVE DOUBLE BAUD RATE CONSTANT.
2250	1146	TAD	BAUDNO	
2251	1375	TAD	(LOBAUD	
2252	3140	DCA	TEMP1	/SETUP
2253	1540	TAD I	TEMP1	
2254	1275	TAD	BITNO	
2255	3140	DCA	TEMP1	
2256	1540	TAD I	TEMP1	
2257	3300	DCA	LORATE	
2258	1276	TAD	TESTNO	/LOWER BAUD LIMIT.
2259	1374	TAD	(JMP I TEST+1	/GET SETTING OF SW'S 1,2,3.
2260	3237	DCA	TEST	/TO DETERMINE TEST TO BE RUN.
2261	0000	TEST,		
2262	1200	TST1		
2263	1200	TST1		/TRANSMIT CONTROL LOGIC TEST.
2264	1400	TST2		/TRANSMIT TIMING TEST.
2265	1600	TST3		/RECEIVE CONTROL LOGIC TEST.
2266	2000	TST4		/RECEIVE TIMING TEST.
2267	2200	TST5		/BREAK TEST.
2268	2400	TST6		/DATA TEST.
2269	2600	TST7		/STATUS REGISTER TEST.

/ROUTINE TO CHECK ALL SWITCHES

2270	3000	DECODE, 0	CLA CLL	
2271	7300	LAS		
2272	7604	DCA	SWITCH	
2273	3274	TAD	SWITCH	
2274	1274	DCA	MASK7	
2275	0773	AND	BAUDNO	/SAVE BAUD RATE SELECTED.
2276	3146	DCA	SWITCH	
2277	1274	TAD		
2278	7012	RTR		
2279	7010	RAR		
2280	0773	AND	MASK7	
2281	3275	DCA	BITNO	/SAVE NO. OF BITS PER DATA WORD.

2264 1274 TAD SWITCH
2265 7012 RTR
2266 7012 RTR
2267 7012 RTR
2270 7012 RTR
2271 0773, MASK7
2272 3276 TESTNO
2273 5650 DECODE
JMP I

0274 0000 SWITCH, 0
0275 0000 BITNO, 0
0276 0000 TESTNO, 0
0277 0000 ONRATE, 0
0300 0000 LORATE, 0
0301 0000 DOUBLE, 0

TABLE OF TEST COUNTER VALUES,

0302 7000 TSTBL, -1000
0303 7000 -1000
0304 7000 -1000
0305 6300 -1500
0306 6000 -2000
0307 5000 -3000
0310 4000 -4000
0311 3000 -5000

ERROR-SCOPE LOOP ROUTINE (FOR TST1 THRU TST5, AND TST7A) //

0312 0000 FAIL,
0313 7344 CLA CLL CMA RAL
0314 1312 TAD FAIL
0315 3140 DCA TEMP1
0316 1372 TAD (7000
0317 3540 DCA I TEMP1
0320 1712 TAD I FAIL
0321 3130 DCA GETBAK
0322 2312 ISE FAIL
0323 1712 TAD I FAIL
0324 3131 DCA NXTST
0325 7004 LAS (4000
0326 0371 AND SNA CLA
0327 7650 JMP I GETBAK
0330 5530 JMP I NXTST
0331 5531

ERROR-SCOPE LOOP ROUTINE FOR TST6 AND TST7B //

0332 0000 SCOPE,
0333 4506 0 UTLS
0334 4474 UKSF
0335 7410 SKP
0336 5344 JMP RCVD
0337 7300 CLA CLL
0340 1142 TAD TOLD
0341 4506 UTLS

0342 7200 CLA
0343 5334 JMP SCOPE+2
0344 4500 UKRB
0345 7604 LAS
0346 0371 AND (4000
0347 7650 SNA CLA
0350 5334 JMP SCOPE+2
0351 5205 JMP BEGIN+5

0371 4000
0372 7000
0373 0466
0374 5640
0375 0063
0376 0053
0377 0400
0400 0400

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/ROUTINE TO SETUP IOT COMMANDS FROM SWITCHES/

SETIOT, 0
0400 0000 LAS
0401 7604 DCA
0402 3140 TAD
0403 1140 AND
0404 0377 RTR
0405 7012 RAR
0406 7010 DCA
0407 3270 TAD
0410 1270 TAD
0411 1376 DCA
0412 3775' TAD
0413 1270 TAD
0414 1374 DCA
0415 3773' TAD
0416 1270 TAD
0417 1372 DCA
0420 3771' TAD
0421 1270 TAD
0422 1370 DCA
0423 3767' TAD
0424 1270 DCA
0425 1366 TAD
0426 3765' DCA
0427 1270 TAD
0430 1364 TAD
0431 3763' DCA
0432 7100 CLL
0433 1140 TAD
0434 0362 AND
0435 7006 RTL
0436 7004 RAL
0437 3267 DCA
0440 1267 TAD
0441 1376 TAD
0442 3761' DCA
0443 1267 TAD

TEMP1
TEMP1
(7700

RCVIOT
RCVIOT
(6000
RKCF
RCVIOT
(6001
RKSF
RCVIOT
(6002
RKCC
RCVIOT
(6004
RKRS
RCVIOT
(6005
RKLE
RCVIOT
(6006
RKRB

TEMP1
(0077

XMTIOT
XMTIOT
(6000
RSPF
XMTIOT

```

2444 1374 TAD (6001
2445 3762' DCA RTSF
2446 1267 TAD XMTIOT
2447 1372 TAD (6002
2450 3757' DCA RTCF
2451 1267 TAD XMTIOT
2452 1370 TAD (6004
2453 3756' DCA RTPC
2454 1267 TAD XMTIOT
2455 1366 TAD (6005
2456 3755' DCA RSPI
2457 1267 TAD XMTIOT
2460 1364 TAD (6006
2461 3754' DCA RPLS
2462 1267 TAD XMTIOT
2463 1353 TAD (6007
2464 3752' DCA RYPK
2465 5600 JMP I SETIOT

```

```

MASK7, 7
XMTIOT, 0
RQVIOT, 0

```

```

0466 0007
0467 0000
0470 0000
0532 0607
0533 6007
0534 0663
0535 0635
0536 0631
0537 0645
0540 0637
0561 0633
0562 0077
0563 0627
0564 6006
0565 0623
0566 6005
0567 0617
0570 6004
0571 0613
0572 6002
0573 0605
0574 6001
0575 0601
0576 6000
0577 7700
0600 0600

```

PAGE

```

0600 0000 XKCF, 0
0601 0000 RKCF, 0
0602 5600 JMP I
0603 7402 HLT
0604 0000 XKSF, 0

```

//IOT SUBROUTINES//
/CLEAR RECEIVE FLAG,

/SKIP ON RECEIVE FLAG;

/CLEAR RECEIVE FLAG AND AC;

/INPUT DATA BUFFER V AC4-11 TO AC4-11;

/DATA 11 TO INTERRUPT ENABLE;
/DATA 10 TO STATUS WORD ENABLE;

/CLEAR AC AND RECEIVE FLAG;
/INPUT DATA BUFFER TO AC4-11;

/SET TRANSMIT FLAG;

/SKIP ON TRANSMIT FLAG;

/CLEAR TRANSMIT FLAG;

/AC4-11 TO OUTPUT DATA BUFFER
/TRANSMIT; SET TRANSMIT FLAG WHEN DONE;

/SKIP ON INTERRUPT REQUEST SET;

2605	0000	RKSF,	0	SKP	
2626	7410		SKP		
2627	2204		ISZ	XKSF	
2610	5604		JMP I	XKSF	
2611	7402		HLT		
2612	0000	XKCC,	0		
2613	0000	RKCC,	0		
2614	5612		JMP I	XKCC	
2615	7402		HLT		
0616	0000	XKRS,	0		
0617	0000	RKRS,	0		
0620	5616		JMP I	XKRS	
0621	7402		HLT		
0622	0000	XK1E,	0		
0623	0000	RK1E,	0		
0624	5622		JMP I	XK1E	
0625	7402		HLT		
0626	0000	XKRB,	0		
0627	0000	RKRB,	0		
0630	5626		JMP I	XKRB	
0631	7402		HLT		
0632	0000	XSPF,	0		
0633	0000	RSPF,	0		
0634	5632		JMP I	XSPF	
0635	7402		HLT		
0636	0000	XTSF,	0		
0637	0000	RTSF,	0		
0640	7410		SKP		
0641	2236		ISZ	XTSF	
0642	5636		JMP I	XTSF	
0643	7402		HLT		
0644	0000	XTCF,	0		
0645	0000	RTCF,	0		
0646	5644		JMP I	XTCF	
0647	7402		HLT		
0650	0000	XTPC,	0		
0651	0000	RTPC,	0		
0652	5650		JMP I	XTPC	
0653	7402		HLT		
0654	0000	XSPI,	0		
0655	0000	RSPI,	0		
0656	7410		SKP		
0657	2254		ISZ	XSPI	
0660	5654		JMP I	XSPI	
0661	7402		HLT		

/CLEAR TRANSMIT FLAG,
/AS4-11 TO OUTPUT DATA BUFFER,
/TRANSMIT, SET TRANSMIT FLAG WHEN DONE.

/GENERATE BREAK,

0662 0000 XTLS, 0
0663 0000 RTLS, 0
0664 5662 JMP I XTLS
0665 7402 HLT

0666 0000 XTPK, 0
0667 0000 RTPK, 0
0670 5666 JMP I XTPK
0671 7402 HLT

1000 PAGE

//ROUTINES TO SETUP ERROR HALT LOCATIONS//

1000 0000 SETT1A, 0
1001 1377 TAD (7402
1002 3776' DCA HALT1
1003 1377 TAD (7402
1004 3775' DCA HALT2
1005 1377 TAD (7402
1006 3774' DCA HALT3
1007 5600 JMP I SETT1A

1010 0000 SETT1B, 0
1011 1377 TAD (7402
1012 3773' DCA HALT4
1013 1377 TAD (7402
1014 3772' DCA HALT5
1015 5610 JMP I SETT1B

1016 0000 SETT1C, 0
1017 1377 TAD (7402
1020 3771' DCA HALT6
1021 1377 TAD (7402
1022 3770' DCA HALT7
1023 5616 JMP I SETT1C

1024 0000 SETT2A, 0
1025 1377 TAD (7402
1026 3767' DCA HALT8
1027 1377 TAD (7402
1030 3766' DCA HALT9
1031 1377 TAD (7402
1032 3765' DCA HALT10
1033 5624 JMP I SETT2A

1034 0000 SETT2B, 0
1035 1377 TAD (7402
1036 3764' DCA HALT11
1037 1377 TAD (7402
1040 3763' DCA HALT12
1041 1377 TAD (7402

1042	3762'	DCA	HALT13
1043	5634	JMP I	SETT2B
1044	0000	SETT3A, 2	
1045	1377	TAD	(7402
1046	3761'	DCA	HALT14
1047	5644	JMP I	SETT3A
1050	0000	SETT3B, 3	
1051	1377	TAD	(7402
1052	3760'	DCA	HALT15
1053	1377	TAD	(7402
1054	3757'	DCA	HALT16
1055	1377	TAD	(7402
1056	3756'	DCA	HALT17
1057	1377	TAD	(7402
1060	3755'	DCA	HALT18
1061	5650	JMP I	SETT3B
1062	0000	SETT3C, 0	
1063	1377	TAD	(7402
1064	3754'	DCA	HALT19
1065	1377	TAD	(7402
1066	3753'	DCA	HALT20
1067	5662	JMP I	SETT3C
1070	0000	SETT3D, 0	
1071	1377	TAD	(7402
1072	3752'	DCA	HALT21
1073	5670	JMP I	SETT3D
1074	0000	SETT4A, 0	
1075	1377	TAD	(7402
1076	3751'	DCA	HALT22
1077	1377	TAD	(7402
1100	3750'	DCA	HALT23
1101	1377	TAD	(7402
1102	3747'	DCA	HALT24
1103	5674	JMP I	SETT4A
1104	0000	SETT5A, 0	
1105	1377	TAD	(7402
1106	3746'	DCA	HALT25
1107	1377	TAD	(7402
1110	3745'	DCA	HALT26
1111	1377	TAD	(7402
1112	3744'	DCA	HALT27
1113	1377	TAD	(7402
1114	3743'	DCA	HALT28
1115	5704	JMP I	SETT5A
1116	0000	SETT7A, 0	
1117	1377	TAD	(7402

1120 3742' DCA HALT33
 1121 1377 TAD (7402
 1122 3741' DCA HALT34
 1123 5716 JMP I SETT7A

1141 2642
 1142 2631
 1143 2256
 1144 2246
 1145 2231
 1146 2215
 1147 2045
 1150 2022
 1151 2012
 1152 1744
 1153 1715
 1154 1703
 1155 1685
 1156 1656
 1157 1643
 1160 1634
 1161 1606
 1162 1513
 1163 1471
 1164 1461
 1165 1434
 1166 1424
 1167 1413
 1170 1310
 1171 1274
 1172 1236
 1173 1242
 1174 1226
 1175 1216
 1176 1207
 1177 7402
 1200

PAGE

//TST1-BASIC TEST OF TRANSMIT LOGIC//

/TST1A=CHECKS THE ABILITY OF-
 /SPF TO SET THE TRANSMIT FLAG;
 /TSF TO SKIP ON TRANSMIT FLAG;
 /CAF TO CLEAR TRANSMIT FLAG;
 /TCF TO CLEAR TRANSMIT FLAG;
 /TSF TO NOT SKIP ON TRANSMIT FLAG EQUAL TO ZERO.

1200	4043	TST1,	SETCNT	/SETUP TEST LOOP COUNTER
1201	4510		SET1A	/SETUP HALT LOCATIONS,
1202	7300	TST1A,	CLA CLL	
1203	4501		USPF	/SET TRANSMIT FLAG;
1204	4502		UTSF	/FLAG SET ?
1205	7610		SKP CLA	

```

1276 5213      JMP      .+5
1277 7402      HALT1,
1278 4544      ERROR
1279 1210      TST1A
1280 1202      TST1AB
1281 1222      CAF
1282 6007      UTSF
1283 4502      JMP      .+5
1284 5222      HALT2,
1285 7402      ERROR
1286 1216      TST1A
1287 1220      TST1AB
1288 1222      USPF
1289 4501      UTSF
1290 1223      TST1B
1291 4503      UTSF
1292 4502      JMP      .+5
1293 5232      HALT3,
1294 7402      ERROR
1295 1227      TST1AB
1296 4544      TST1B=2
1297 1222      ISZ
1298 1231      JMP
1299 1232      TSTCNT
1300 5202      TST1A

```

/TST1B-CHECKS THAT NO OTHER DEVICE WILL CAUSE AN INTERRUPT;
/AND THAT TRANSMIT FLAG WILL CAUSE AN INTERRUPT.

```

1234 4043      SETCNT
1235 4511      SET1B
1236 7300      CLA CLL
1237 6007      CAF
1238 4505      USPI
1239 5246      JMP      .+5
1240 7402      HALT4,
1241 4544      ERROR
1242 1243      TST1B
1243 1236      TST1BB
1244 1246      CLA CLL
1245 7300      CAF
1246 6007      TAD
1247 1377      DCA
1248 3002      USPF
1249 4501      ION
1250 6001      NOP
1251 7000      IOF
1252 6002      HALT5,
1253 7402      ERROR
1254 4544      TST1BB
1255 1246      TST1C=2
1256 1264      ISZ
1257 2132      JMP
1258 5236      TSTCNT
1259      TST1B

```

/TST1C-CHECKS THAT SPI WILL SKIP ON INTERRUPT REQUEST
/AND THAT INTERRUPT ENABLE ALONE WILL NOT CAUSE AN INTERRUPT.

```

//KL8F DOUBLE BUFFERED ASYNCHRONOUS INTERFACE PAL10 V141 22-MAY-72 13157 PAGE 1-14

1264 4043 SETCNT
1265 4512 SET1C
1266 7300 CLA CLL
1267 6007 CAP
1270 4531 USPF
1271 4505 USPI
1272 7410 SKP
1273 5300 JMP
1274 7432 HALT6,
1275 4544
1276 1266
1277 1300
1300 7300 TST1CB, CLA CLL
1301 1376 TAD
1302 3002 DCA
1303 6007 CAP
1304 6001 ION
1305 7000 NOP
1306 6002 IOF
1307 5314 JMP
1310 7402 HALT7,
1311 4544 ERROR
1312 1300 TST1CB
1313 1316 CHECK1
1314 2132 ISZ
1315 5266 JMP
1316 7200 CHECK1, CLA
1317 1375 TAD
1320 7421 MQL
1321 1774, TAD
1322 7640 SZA CLA
1323 5200 JMP
1324 5773, JMP
1373 1400
1374 0276
1375 0001
1376 1310
1377 1202 1400

//TST2=TRANSMIT TIMING TEST//
//TST2=CHECKS THE ABILITY OF-
//TPC TO SET THE TRANSMIT FLAG,
//TLS TO CLEAR THE TRANSMIT FLAG,
//TLS TO SET THE TRANSMIT FLAG,

1400 4043 TST2, SETCNT
1401 4513 SET2A
1402 7300 CLA CLL
1403 1777, TAD DOUBLE

1403 1777, TAD DOUBLE
//SETUP TEST LOOP COUNTER,
//SETUP HALT LOCATIONS,
//CLEAR ALL FLAGS,
//SET TRANSMIT FLAG,
//SKIP ON INTERRUPT REQUEST SET,
//INTERRUPT REQ, SET, CONTINUE,
//SPI FAILED OR INT, REQ, NOT SET,
//SCOPE LOOP, PRESS CONTINUE TO ENTER,
//LOOP ADDRESS,
//NEXT TEST,
//GET RETURN ADDRESS,
//SETUP RETURN,
//CLEAR ALL FLAGS,
//INTERRUPT ON,
//WAIT
//INTERRUPT OFF,
//NO INTERRUPT, CONTINUE,
//ILLEGAL INTERRUPT,
//SCOPE LOOP PRESS CONTINUE TO ENTER,
//LOOP ADDRESS,
//NEXT TEST
//TEST LOOP COUNTER,
//LOOP,
//LOAD CURRENT TEST
//NUMBER INTO HQ REGISTER,
//ARE ALL TESTS BEING RUN?
//YES, GO TO NEXT TEST,
//NO, CONTINUE CURRENT TEST,
//NXT TEST,
//SETUP TEST LOOP COUNTER,
//SETUP HALT LOCATIONS,
//SET DELAY ROUTINE TO DELAY TWICE AS

```

```

1404 3124 DCA NDELAY
1405 4503 UTCF
1406 4504 UTPC
1407 4020 DELAY
1410 4502 UTSF
1411 7410 SKP
1412 5217 JMP
1413 7402 HALT8,
1414 4544 HLT
1415 1402 ERROR
1416 1417 TST2A
1417 7300 TST2AB, CLA CLL
1420 4501 USPF
1421 4506 UTLS
1422 4502 UTSF
1423 5230 JMP
1424 7402 HALT9,
1425 4544 HLT
1426 1417 ERROR
1427 1442 TST2AB
1430 4020 TST2B=2
1431 4502 DELAY
1432 7410 UTSF
1433 5240 SKP
1434 7402 JMP
1435 4544 HALT10,
1436 1417 HLT
1437 1442 ERROR
1440 2132 TST2B=2
1441 5202 ISE
1442 5202 JMP
1443 5202 TSTCNT
1444 5202 TST2A

```

/TST2B - CHECKS THAT THE TRANSMIT FLAG SETS WITHIN
/THE PROPER TIME FOR THE BAUD RATE SELECTED.

```

1442 4043 SETCNT
1443 4514 SET2B
1444 7300 CLA CLL
1445 1776 TAD
1446 3124 LORATE
1447 4506 NDELAY
1450 4502 DCA
1451 5250 UTL
1452 4506 UTSF
1453 4502 JMP
1454 5253 UTL
1455 4506 UTSF
1456 4020 DELAY
1457 4502 UTSF
1460 5265 JMP
1461 7402 HALT11,
1462 4544 HLT
1463 1444 ERROR
1464 1475 TST2B
1465 4020 DELAY

```

/SETUP TEST LOOP COUNTER,
/SETUP HALT LOCATIONS,
/SET DELAY TO LOWER BAUD LIMIT,
/INITIALIZE TRANSMIT FLAG,
/FOR UPCOMING TIMING TEST,

/WAIT, STILL CLEAR?
/FLAG, CONTINUE?
/YES, CONTINUE?
/NO, FLAG SET TOO SOON,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,
/LOOP ADDRESS,
/NEXT TEST,
/WAIT AGAIN, TO INSURE THAT FLAG WILL SET,


```

1636      7402      HALT14, HLT
1637      4544      ERROR
1638      1610      TST3A
1639      1611      TST3B=2
1640      1612      ISZ
1641      1613      JMP
1642      2132      TSTCNT
1643      5202      TST3A
1644      /NO KCC FAILED
1645      /SCOPE LOOP, PRESS CONTINUE TO ENTER,
1646      /LOOP ADDRESS,
1647      /NEXT TEST,
1648      /TEST LOOP COUNTER,
1649      /LOOP,

```

```

1614 /TST3B - CHECKS THE ABILITY OF-
1615 /LS TO SET THE RECEIVE FLAG,
1616 /KSF TO SKIP ON RECEIVE FLAG EQUAL TO ONE,
1617 /KCF TO CLEAR RECEIVE FLAG,
1620 /KCC TO CLEAR RECEIVE FLAG;
1621 /KSF TO NOT SKIP ON RECEIVE FLAG EQUAL TO ZERO;
                                /SETUP TEST COUNTER,
                                /SETUP HALT LOCATIONS;
                                /SET DELAY ROUTINE TO DELAY TWICE AS
                                /LONG AS TIME NEEDED;
                                /CLEAR ALL FLAGS.

1614 4043
1615 4516
1616 7300
1617 1777,
1620 3124
1621 6007

```

1625	UTLS	/TRANSMIT
1626	DELAY	/DELAY TWICE MAX TIME;
1627	UTSF	/TRANSMIT FLAG SET;
1630	JMP	/WAIT FOR IT
1631	UKSF	/NOW RECEIVE FLAG SHOULD ALSO BE SET;
1632	SKP	
1633	JMP	/RECEIVE FLAG SET, CONTINUE;
1634	HLT	/RECEIVE FLAG OR KSF FAILED;
1635	ERROR	/SCOPE LOOP, PRESS CONTINUE TO ENTER;
1636	TST3B	/LOOP ADDRESS
1637	TST3B8	/NEXT TEST;
1640	UKCC	/CLEAR RECEIVE FLAG;
1641	UKSF	/SKIP ON RECEIVE FLAG=1;
1642	JMP	/RECEIVE FLAG CLEAR CONTINUE;
1643	HLT	/KCC OR KSF FAILED;
1644	ERROR	/SCOPE LOOP, PRESS CONTINUE TO ENTER;
1645	TST3B	/LOOP ADDRESS;
1646	TST3B8	/NEXT TEST;

```

0077      /CLEAR RECEIVE FLAG,  

0080      /TRANSMIT,  

0083      /DELAY TWICE MAX TIME;  

0086      /SKIP ON RECEIVE FLAG;  

0089      /FLAG SET CONTINUE  

0092      /RECEIVE FLAG OR KSF FAILED,  

0095      /SCOPE LOOP, PRESS CONTINUE TO ENTER,  

0098      /LOOP ADDRESS,  

0101      /NEXT TEST,  

0104      /CLEAR RECEIVE FLAG,  

0107      /RECEIVE FLAG CLEARED?  

0110      /YES, CONTINUE  


```

```

1665 7402      HALT18, HLT
1666 4544      ERROR
1667 1647      TST3B8
1670 1673      TST3C=2
1671 2132      ISZ
1672 5216      JMP
          TSTCNT
          TST3B

```

```

1673 4043      /TST3C - CHECKS THE ABILITY OF-
1674 4517      /KIE TO DISABLE INTERRUPT ENABLE F/F,
1675 7300      /KIE TO ENABLE INTERRUPT ENABLE F/F,
1676 6007      SETCNT
1677 4477      /SETUP TEST LOOP COUNTER,
1678 4501      /SETUP HALT LOCATIONS,
1679 4505      /CLEAR ALL FLAGS, ENABLE INT, ENABLE F/F,
1680 5307      /DISABLE INTERRUPT ENABLE F/F,
1681 7402      /SET TRANSMIT FLAG,
1682 4544      /INTERRUPT REQUEST SET?
1683 4501      /NO, CONTINUE.
1684 4505      /YES, KIE FAILED.
1685 1675      /SCOPE LOOP, PRESS CONTINUE TO ENTER,
1686 1707      /LOOP ADDRESS,
1687 7201      /NEXT TEST,
1688 4477      /AC11=1
1689 4501      /ENABLE INTERRUPT ENABLE F/F
1690 4505      /SET TRANSMIT FLAG
1691 5307      /INTERRUPT REQUEST SET?
1692 5321      /YES, CONTINUE
1693 7402      /NO, KIE FAILED,
1694 4544      /SCOPE LOOP, PRESS CONTINUE TO ENTER
1695 1707      /LOOP ADDRESS,
1696 1725      /NEXT TEST,
1697 2132      /TEST LOOP COUNT,
1698 5275      /LOOP,
          TSTCNT
          TST3C

```

/TST3D - CHECKS THAT THE RECEIVE FLAG WILL CAUSE AN INTERRUPT,

```

1723 4043      SETCNT
1724 4520      SET3D
1725 7300      CLA CLL
1726 6007      CAF
1727 1376      TAD
1730 3002      DCA
1731 4506      UTLS
1732 4502      UTSP
1733 5332      JMP
1734 4506      UTLS
1735 4020      DELAY
1736 4502      UTSP
1737 5336      JMP
1740 4503      ION
1741 6001      NOP
1742 7000      NOP
          (RINTOK
          RETURN
          .-1
          .-1
          /TRANSMIT,
          /DELAY TWICE MAX TIME,
          /WAIT FOR TRANSMIT FLAG, AT WHICH TIME
          /RECEIVE FLAG SHOULD ALSO BE UP,
          /CLEAR TRANSMIT FLAG
          /TURN INTERRUPT ON,
          /WAIT

```

```

1743 6002 IOF
1744 7402 HALT21, HLT
1745 4544 ERROR
1746 1725 TST3D
1747 1752 CHECK3
1750 2132 RINTOK, ISZ
1751 5325 JMP
1752 7200 CHECK3, CLA
1753 1375 TAD
1754 7421 MQL
1755 1774, TAD
1756 7640 SZA CLA
1757 5200 JMP
1760 5773, JMP

1773 2000
1774 0276
1775 0003
1776 1750
1777 0301
2000

```

PAGE

//TST4 - RECEIVER TIMING TEST//

//TST4A - CHECKS THE ABILITY OF THE RECEIVER FLAG TO BE SET,
//THAT IT WILL NOT SET TOO SOON, AND THAT IT WILL COME UP
//WITHIN THE TIME SPECIFIED BY THE BAUD RATE SELECTED.

```

2000 4043 TST4, SETCNT
2001 4521 SET4A
2002 7300 CLA CLL
2003 1777, TAD LORATE
2004 3124 DCA NDELAY
2005 4475 UKCC
2006 4506 UTLS
2007 4020 DELAY
2010 4474 UKSF
2011 5216 JMP
2012 7402 HALT22, HLT
2013 4544 ERROR
2014 2002 TST4A
2015 2026 TST4AB
2016 4020 DELAY
2017 4474 UKSF
2020 7410 SKP
2021 5226 JMP
2022 7402 HALT23, HLT
2023 4544 ERROR
2024 2002 TST4A
2025 2026 TST4AB

2026 7300 TST4AB, CLA CLL
2027 1776, TAD ONRATE

```

//SET DELAY ROUTINE TO WAIT THE

2216	4544	ERROR		/SCOPE LOOP, PRESS CONTINUE TO ENTER,
2217	2204	TST5A		/LOOP ADDRESS,
2220	2221	TST5AB		/NEXT TEST,
2221	7300	TST5AB, CLA CLL	LOBKCT	
2222	1274	TAD	NDELAY	
2223	3124	DCA		/GET LOWER TIME LIMIT,
2224	4503	UTCF		/SAVE IT,
2225	4507	UTPK		/CLEAR TRANSMIT FLAG,
2226	4020	DELAY		/GENERATE BREAK,
2227	4502	UTSF		/WAIT,
2230	5235	JMP	.+5	/TRANSMIT FLAG SET?
2231	7402	HLT		/NO, CONTINUE,
2232	4544	ERROR		/YES, FLAG SET TOO SOON,
2233	2221	TST5AB		/SCOPE LOOP, PRESS CONTINUE TO ENTER,
2234	2235	TST5AC		/LOOP ADDRESS,
				/NEXT TEST,
2235	7300	TST5AC, CLA CLL	BRKCT	
2236	1275	TAD	NDELAY	
2237	3124	DCA		/SET DELAY ROUTINE TO WAIT THE
2240	4503	UTCF		/EXACT TIME REQUIRED TO SET THE FLAG,
2241	4507	UTPK		/CLEAR TRANSMIT FLAG,
2242	4020	DELAY		/GENERATE BREAK,
2243	4502	UTSF		/WAIT,
2244	7410	SKP		/TRANSMIT FLAG SET?
2245	5252	JMP	.+5	
2246	7402	HLT		/YES, CONTINUE,
2247	4544	ERROR		/NO, FLAG NOT SET OR SET TOO LATE,
2250	2235	TST5AC		/SCOPE LOOP, PRESS CONTINUE TO ENTER,
2251	2235	TST5AD		/LOOP ADDRESS,
2252	4503	UTCF		/NEXT TEST,
2253	4020	DELAY		/CLEAR TRANSMIT AND BRK FLAGS,
2254	4502	UTSF		
2255	5262	JMP	.+5	/TRANSMIT FLAG SET?
2256	7402	HLT		/NO, CONTINUE,
2257	4544	ERROR		/YES, TRANSMIT AND/OR BRK FLAG STILL SET,
2260	2235	TST5AC		/SCOPE LOOP, PRESS CONTINUE TO ENTER,
2261	2264	CHECK5		/LOOP ADDRESS,
2262	2132	ISE		/NEXT TEST,
2263	5204	JMP	TSTCNT	/TEST LOOP COUNTER,
2264	7200	CLA	TST5A	/LOOP,
2265	1376	TAD	(5	/LOAD CURRENT TEST
2266	7421	MQL		/NUMBER INTO HQ REGISTER,
2267	1775	TAD	TESTNO	/ARE ALL TESTS BEING RUN?
2270	7640	SZA CLA		/YES, GO TO NEXT TEST,
2271	5202	JMP	TST5A=2	/NO, CONTINUE CURRENT TEST,
2272	5774	JMP	TST6	/NXT TEST,

2273	1000	HIBKCT,	-7000
2274	5060	LOBKCT,	-2720
2275	2564	BRKCT,	-5214
2374	2400		
2375	0276		
2376	0005		

2377 7773
2400

PAGE

//TST6 - DATA TEST//

/TST6A - BINARY COUNT PATIERN,

2400	4043	TST6,	SETCNT	/SETUP TEST COUNTER,
2401	4475	TST6A,	UKCC	/CLEAR AC AND RECEIVE FLAG,
2402	3133		DCA	/INITIALIZE TRANSMIT AND RECEIVE,
2403	3142		DCA	/DATA HOLDING LOCATIONS,
2404	3143		DCA	
2405	4506		UTLS	/TRANSMIT,
2406	4474	LOOP6A,	UKSF	/IS RECEIVE FLAG SET?
2407	7410		SKP	/NO, CHECK TRANSMIT,
2410	5226		JMP	/YES, GO AND CHECK DATA RECEIVED,
2411	4502		UTSF	/IS TRANSMIT FLAG SET?
2412	5206		JMP	/NO, CHECK FLAGS AGAIN,
2413	7300		CLA CLL	
2414	1143		TAD	/GET LAST CHARACTER TRANSMITTED,
2415	3142		DCA	/SAVE IT FOR COMPARISON,
2416	2133		ISZ	/SETUP NEXT CHARACTER,
2417	7000		NOP	
2420	1133		TAD	/SAVE IT,
2421	3143		DCA	/TRANSMIT NEXT CHARACTER,
2422	1143		TAD	
2423	4506		UTLS	
2424	7200		CLA	
2425	5206	RCV6A,	JMP	
2426	4500		UKRB	/RECEIVE,
2427	0134		AND	/MASK OUT UNWANTED BITS,
2430	3140		DCA	/SAVE DATA,
2431	1140		TAD	/GET IT BACK FOR COMPARISON,
2432	7041		CIA	
2433	3137		DCA	/SAVE COMPLEMENT,
2434	1142		TAD	/COMPARE TRANSMIT AND RECEIVE DATA,
2435	0134		AND	/MASK OUT UNWANTED BITS,
2436	1137		TAD	/GET COMPLEMENT AND COMPARE,
2437	7650		SNA CLA	/DOES TRANSMIT = RECEIVE?
2440	5250		JMP	/YES, CONTINUE,
2441	1140		TAD	/BAD DATA TO AC,
2442	7402	HALT29,	HLT	
2443	7200		CLA	/GOOD DATA TO AC,
2444	1142		TAD	/MASK OUT UNWANTED BITS,
2445	0134		AND	
2446	7402	HALT30,	HLT	
2447	4545		DATERR	/SCOPE LOOP, PRESS CONTINUE TO ENTER,
2450	2132	UPDATE,	ISZ	/TEST LOOP COUNTER,
2451	5206		JMP	/LOOP
2452	4474		UKSF	/WAIT FOR RECEIVE FLAG
2453	5252		JMP	/TO SET, BEFORE GOING ON,

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/TST68 - RANDOM DATA PATTERN,

```
2454 4043 TST68, SETCNT (1233 /SETUP TEST COUNTER,
2455 1377 TAD RP1B /SETUP RANDOM NUMBER
2456 3135 DCA RP1B /GENERATOR,
2457 1376 TAD (7622
2460 3136 DCA RP2B
2461 4475 UKCC RAN1
2462 4324 JMS
2463 4506 UTLS
2464 4474 UKSF
2465 7410 SKP RCV68
2466 5275 JMP UTSF
2467 4502 UTSF
2470 5264 JMP LOOP68
2471 4324 JMS RAN1
2472 4506 UTLS
2473 7200 CLA
2474 5264 JMP LOOP68
2475 4500 UKRB
2476 0134 AND BITMSK
2477 3140 DCA TEMP1
2480 1140 TAD TEMP1
2481 7041 CIA
2482 1142 TAD TOLD
2483 7650 SNA CLA
2484 5313 JMP
2485 1140 TAD
2486 7402 HLT
2487 7200 CLA
2488 1142 TAD
2489 7402 HLT
2490 4545 DATERR
2491 2132 ISE
2492 5264 JMP
2493 7200 CLA
2494 1375 TAD
2495 7421 MQL
2496 1774 TAD
2497 7640 SEA CLA
2498 5200 JMP
2499 5773 JMP
```

//RANDOM NUMBER GENERATORS//

```
2524 0000 RAN1, 0
2525 7300 CLA CLL
2526 1142 TAD
2527 3143 DCA
2530 1350 TAD
2531 3142 DCA
2532 1135 TAD
2533 7006 RTL
2534 1136 TAD
2535 3135 DCA
```

```
2536 1135 TAD RP13
2537 7006 RTL
2540 1136 TAD RP28
2541 7006 RTL
2542 3136 DCA
2543 1135 TAD RP18
2544 0134 AND BITMSK
2545 3350 DCA TNEW
2546 1350 TAD TNEW
2547 5724 JMP 1
2550 0000 TNEW, 0
```

/MASK OUT UNWANTED BITS.

PAGE

//TST7 - STATUS REGISTER TEST//

//TST7A - FORCES AN OVERRUN ERROR AND CHECKS THAT THIS CONDITION
//WILL CAUSE THE OVERRUN ERROR BIT (AC02) TO COME UP, AND THAT
//SWE AND OVERRUN WILL CAUSE AC00 TO COME UP.

```
2600 4043 TST7, SETCNT
2601 4523 SET7A
2602 4475 TST7A, UKCC
2603 1377 TAD (0002
2604 4477 UKIE
2605 4754 JMS 1
2606 3140 DCA RANGEN
2607 1140 TAD TEMPI
2610 4506 LOOP7A, TAD UTLS
2611 4502 UTLSF
2612 5211 JMP 1
2613 4506 UTLS
2614 4502 UTLSF
2615 5214 JMP 1
2616 4506 UTLS
2617 4502 UTLSF
2620 5217 JMP 1
2621 4500 UKRB
2622 0376 AND (5000
2623 3137 DCA HOLD1
2624 1137 TAD HOLD1
2625 0375 AND (1000
2626 7640 SZA CLA
2627 7410 SKP
2630 5235 JMP 1
2631 7402 HALT33, HLT
2632 4544 ERROR
```

/SETUP TEST COUNTER,
/SETUP HALT LOCATIONS,
/CLEAR AC AND RECEIVE FLAG,
/ENABLE SWE,

/GET RANDOM DATA,
/SAVE RANDOM NUMBER JUST
/IN CASE IT IS NEEDED FOR SCOPE LOOP,
/TRANSMIT!
/SKIP ON TRANSMIT FLAG,
/REPEAT TRANSMIT TWICE MORE
/TO FORCE OVERRUN,

/RECEIVE AND READ ERROR BITS,
/MASK FOR ANY ERROR AND OVERRUN,
/SAVE ERROR BITS,
/MASK FOR OVERRUN ERROR BIT,
/OVERRUN ERROR=1?

/YES, CONTINUE,
/FAILED, OVERRUN ERROR DID NOT OCCUR,
/SCOPE LOOP, PRESS CONTINUE TO ENTER,

2714 7640 SZA CLA
2715 7410 SKP
2716 5322 JMP
2717 1140 TAD

I*4
TEMP1

/FAILING STATUS BIT AND DATA
/WORD TO AC.

HALT36, HLT

2720 7402

DATERR

2721 4545

TAD

2722 1140

TEMP1

2723 0372

(0400

2724 7640

SZA CLA

2725 7410

SKP

2726 5332

I*4

2727 1140

TEMP1

2728 7402

DATERR

2729 4545

TEMP1

2730 7402

(0400

2731 1140

SZA CLA

2732 0374

AND

2733 7640

SKP

2734 7410

I*4

2735 5342

TEMP1

2736 1140

TAD

2737 7402

DATERR

2738 4545

TEMP1

2739 2132

TSYCN

2740 7402

LOOP7B

2741 5260

JMP

2742 7200

CLA

2743 1371

(7

2744 7421

TESTNO

2745 1770

TSY7

2746 7640

TSY1

2747 5200

TSY1

2748 4020

TSY1

2749 5767

JMP

2750 2524

RANGEN, RAN1

2751 1200

2752 0276

2753 0007

2754 0400

2755 2000

2756 4000

2757 1000

2758 5000

2759 0002

2760 3000

PAGE

/EXACT BAUD RATE TABLES
/USED WHEN TESTING FLAGS TO SET WITHIN SPECS

/110 BAUD

/7 BIT TIME

/8 BIT TIME

/9 BIT TIME

ON110, -1234

-1373

-1533

3000 6544

3001 6405

3002 6245

3003 6106	-1672	/10 BIT TIME
3004 5746	-2032	/11 BIT TIME
3005 5606	-2172	/12 BIT TIME
3006 6737	ON134,	/134.5 BAUD
3007 6620	-1041	/7 BIT TIME
3010 6502	-1160	/8 BIT TIME
3011 6364	-1276	/9 BIT TIME
3012 6246	-1414	/10 BIT TIME
3013 6127	-1532	/11 BIT TIME
	-1651	/12 BIT TIME
3014 7027	ON150,	/150 BAUD
3015 6720	-751	/7 BIT TIME
3016 6613	-1060	/8 BIT TIME
3017 6505	-1165	/9 BIT TIME
3020 6376	-1273	/10 BIT TIME
3021 6271	-1402	/11 BIT TIME
	-1507	/12 BIT TIME
3022 7413	ON300,	/7 BIT TIME
3023 7351	-365	/8 BIT TIME
3024 7305	-427	/9 BIT TIME
3025 7242	-473	/10 BIT TIME
3026 7200	-536	/11 BIT TIME
3027 7134	-600	/12 BIT TIME
	-644	
3030 7606	ON600,	/600 BAUD
3031 7564	-172	/7 BIT TIME
3032 7544	-214	/8 BIT TIME
3033 7522	-234	/9 BIT TIME
3034 7500	-256	/10 BIT TIME
3035 7457	-300	/11 BIT TIME
	-321	/12 BIT TIME
3036 7703	ON1200,	/1200 BAUD
3037 7672	-75	/7 BIT TIME
3040 7661	-106	/8 BIT TIME
3041 7651	-117	/9 BIT TIME
3042 7637	-127	/10 BIT TIME
3043 7627	-141	/11 BIT TIME
	-151	/12 BIT TIME
3044 7727	ON1800,	/1800 BAUD
3045 7722	-51	/7 BIT TIME
3046 7713	-56	/8 BIT TIME
3047 7705	-65	/9 BIT TIME
3050 7700	-73	/10 BIT TIME
3051 7672	-100	/11 BIT TIME
	-106	/12 BIT TIME
3052 7742	ON2400,	/2400 BAUD
3053 7735	-36	/7 BIT TIME
	-43	/8 BIT TIME

3054	7731	-47	/9 BIT TIME
3055	7724	-54	/10 BIT TIME
3056	7720	-60	/11 BIT TIME
3057	7713	-65	/12 BIT TIME

LOW BAUD RATE TABLES
USED WHEN TESTING FLAGS NOT TO SET TOO SOON

3060	7003	L0110,	-775	/110 BAUD
3061	6672		-1106	/7 BIT TIME
3062	6562		-1216	/8 BIT TIME
3063	6451		-1327	/9 BIT TIME
3064	6340		-1440	/10 BIT TIME
3065	6227		-1551	/11 BIT TIME
				/12 BIT TIME

3066	7141	L0134,	-637	/134.5 BAUD
3067	7045		-733	/7 BIT TIME
3070	6731		-1027	/8 BIT TIME
3071	6656		-1122	/9 BIT TIME
3072	6562		-1216	/10 BIT TIME
3073	6466		-1312	/11 BIT TIME
				/12 BIT TIME

3074	7213	L0150,	-565	/150 BAUD
3075	7126		-652	/7 BIT TIME
3076	7041		-737	/8 BIT TIME
3077	6793		-1029	/9 BIT TIME
3100	6666		-1112	/10 BIT TIME
3101	6601		-1177	/11 BIT TIME
				/12 BIT TIME

3102	7505	L0300,	-273	/300 BAUD
3103	7453		-325	/7 BIT TIME
3104	7420		-360	/8 BIT TIME
3105	7366		-412	/9 BIT TIME
3106	7333		-445	/10 BIT TIME
3107	7300		-500	/11 BIT TIME
				/12 BIT TIME

3110	7643	L0600,	-135	/600 BAUD
3111	7626		-152	/7 BIT TIME
3112	7611		-167	/8 BIT TIME
3113	7573		-205	/9 BIT TIME
3114	7556		-222	/10 BIT TIME
3115	7541		-237	/11 BIT TIME
				/12 BIT TIME

3116	7722	L01200,	-56	/1200 BAUD
3117	7712		-66	/7 BIT TIME
3120	7704		-74	/8 BIT TIME
3121	7676		-102	/9 BIT TIME
				/10 BIT TIME

3122	7666	-112	/11 BIT TIME
3123	7660	-120	/12 BIT TIME
3124	7741	L01800, -37	/1800 BAUD
3125	7735	-43	/7 BIT TIME
3126	7730	-50	/8 BIT TIME
3127	7723	-55	/9 BIT TIME
3130	7717	-61	/10 BIT TIME
3131	7712	-66	/11 BIT TIME
3132	7751	L02400, -27	/2400 BAUD
3133	7746	-32	/7 BIT TIME
3134	7742	-36	/8 BIT TIME
3135	7736	-42	/9 BIT TIME
3136	7733	-45	/10 BIT TIME
3137	7730	-50	/11 BIT TIME

S

0177 0302

4000
4100

4200
4300

4400
4500

4600
4700

5000
5100

5200
5300

5400
5500

5600
5700

6000
6100

6200
6300

6400
6500

6600
6700

7000
7100

7200
7300

7400
7500

7600
7700

BAUDNO	0146	HALT36	2720	RKIE	0623	TST188	1246
BEGIN	3200	HALT37	2730	RKRB	0627	TST1C	1266
BITMSK	0134	HALT38	2740	RKRS	0617	TST1CB	1300
BITNO	0275	HALT4	1242	RKSF	0605	TST2	1400
BRKCN	2275	HALT5	1256	RP1B	0135	TST2A	1402
CAF	6007	HALT6	1274	RP2B	0136	TST2AB	1417
CHECK1	1316	HALT7	1310	RSPF	0633	TST2B	1444
CHECK2	1521	HALT8	1413	RSPI	0655	TST28	1475
CHECK3	1752	HALT9	1424	RTCF	0645	TST28B	1600
CHECK4	2053	H1BKCT	2273	RTLS	0663	TST3	1602
CHECK5	2264	HOLD1	0137	RTPC	0651	TST3A	1616
CON100	0126	INTOK	1262	RTPK	0667	TST3B	1647
COUNT	0043	LO110	3060	RTSF	0637	TST38B	1675
DATERR	4545	LO1200	3116	SCOPE	0332	TST3C	1707
DECODE	0250	LO134	3066	SET1A	4510	TST3CB	1725
DELAY	4020	LO150	3074	SET1B	4511	TST3D	2000
DELLAY	0125	LO1600	3124	SET1C	4512	TST4	2002
DELOOP	0024	LO2400	3132	SET2A	4513	TST4A	2026
DOUBLE	0301	LO300	3102	SET2B	4514	TST5	2200
ERROR	4544	LO600	3110	SET3A	4515	TST5A	2204
FAIL	0312	LOBAUQ	0063	SET3B	4516	TST5AB	2221
GETBAK	0130	LOBKCT	2274	SET3C	4517	TST5AC	2235
HALT1	1207	LOOP6A	2406	SET3D	4520	TST5AD	2252
HALT10	1434	LOOP6B	2464	SET4A	4521	TST6	2400
HALT11	1461	LOOP7A	2607	SET5A	4522	TST6A	2401
HALT12	1471	LOOP7B	2660	SET7A	4523	TST6B	2454
HALT13	1513	LORATE	0300	SETCNT	4043	TST7	2600
HALT14	1606	MASK7	0466	SETIOT	0400	TST7A	2602
HALT15	1634	NOL	7421	SETT1A	1000	TST7B	2650
HALT16	1643	NDELAY	0124	SETT1B	1010	TSTBL	0302
HALT17	1656	NXTST	0131	SETT1C	1016	TSTCNT	0132
HALT18	1665	ON110	3000	SETT12A	1024	UKCF	4475
HALT19	1703	ON1200	3036	SETT12B	1034	UKCF	4473
HALT2	1216	ON134	3006	SETT13A	1044	UKIE	4477
HALT20	1715	ON150	3014	SETT13B	1050	UKRB	4500
HALT21	1744	ON1600	3044	SETT13C	1062	UKRS	4500
HALT22	2012	ON2400	3052	SETT13D	1070	UKSF	4474
HALT23	2022	ON300	3022	SETT14A	1074	UPDATE	2450
HALT24	2045	ON600	3030	SETT15A	1104	US100	0127
HALT25	2215	ONBAUD	0053	SETT17A	1116	USPF	4501
HALT26	2231	ONRATE	0277	SETT17B	0274	USPI	4505
HALT27	2246	RAN1	2524	SWITCH	0140	UTCF	4503
HALT28	2256	RANGEN	2754	TEST1	0237	UTLS	4506
HALT29	2442	RCV6A	2426	TESTNO	0276	UTPC	4504
HALT3	1226	RCV6B	2475	TLAST	0143	UTPK	4507
HALT30	2446	RCV7B	2700	TNEW	2550	UTSF	4502
HALT31	2506	RCVD	0344	TOLD	0141	XPALL	0144
HALT32	2511	RCV10T	0470	TST1	1200	XKCC	0612
HALT33	2631	RETURN	0002	TST1A	1202	XKCF	0600
HALT34	2642	RINTOK	1750	TST1B	1222	XKIE	0622
HALT35	2710	RKCC	0613	TST1B	1236	XKRB	0626
		RKCF	0601			XKRS	0616

X4SF 2604
X4TCH 0133
X4Y10T 0467
XSCOPE 0145
XSET1A 0110
XSET1B 0111
XSET1C 0112
XSET1A 0113
XSET2B 0114
XSET3A 0115
XSET3B 0116
XSET3C 0117
XSET3D 0120
XSET4A 0121
XSET5A 0122
XSET7A 0123
XSPF 0632
XSPI 0654
XTCF 0644
XTLS 0662
XTPC 0650
XTPK 0666
XTSF 0636
XXKCC 0075
XXKCF 0073
XXKIE 0077
XXKRB 0100
XXKRS 0076
XXKSF 0074
XXSPF 0101
XXSPI 0105
XXTCF 0103
XXTLS 0106
XXTPC 0104
XXTPK 0107
XXTSF 0102

ERRORS DETECTED: 0

LINKS GENERATED: 67

RUN-TIME: 13 SECONDS

3K CORE USED